

ChE 427 (Engineering and Entrepreneurship)

Department of Chemical Engineering

University of Illinois Chicago

Fall 2025

Instructors:

Said Al-Hallaj, PhD

Course Time: Wednesday 3:30 to 6:15 pm

Classroom (in person): EIB 271 (945 W Taylor Street, Chicago, IL 60607)

Office Hours: By appointment

Prerequisite: course suitable for senior and graduate engineering students.

Credit Hours: 4 hr (2-credit for lecture and 2-credit for case study)

Course Description:

The course provides an introductory view into the various aspects involved in technology commercialization (i.e. taking an idea from the lab to the marketplace). Following is an outline of the course outline:

Part I: Intellectual Property

- 1) Confidentiality of Non-Disclosure Agreement (NDA)
- 2) Patent applications
- 3) Copyrights
- 4) Know-how

Part II: UIC Resources

- 1) Office of Technology Management
- 2) Invention Center
- 3) Business Plan Competition
- 4) mHUB
- 5) Illinois Venture

Part III: Business Development Strategy

- 1) Business Plan
- 2) Marketing strategies
- 3) Finance and accounting

Part IV: Starting a Business

- 1) Legal structure (LLC, Inc., etc)
- 2) Operating Agreement
- 3) Governance

Part V: Fundraising

- 1) Grants
- 2) Seed/Angel investment
- 3) Venture capital
- 4) Strategic partnerships

Grade:

25% Weekly assignments (3-4 mini-projects, quizzes or written essays)

25% Midterm Home Exam

50% Final Project: written plan (25%) and oral presentation (25%)

Final Project: technology commercialization strategy (i.e. IP assessment, market study, go-to-market strategy)

Class Schedule

<u>Week</u>	<u>Topic</u>
8/27	(1) Introduction (Anderson Entrepreneurship Quiz)
9/3	(2) Protecting your intellectual property (IP) (Guest Lecturer: UIC-OTM)
9/10	(3) How Universities commercialize technologies (Guest Lecturer: UIC-OTM)
9/17	(4) Final Project Proposal
9/24	(5) What Makes Entrepreneurs Entrepreneurial
10/1	(6) Pitching Your Idea <u>(Project Abstract Due)</u>
10/8	(7) Why Marketing Study is Important
10/15	(8) Financials Basics
10/22	(9) Starting Your Own Business (Entrepreneurial Death Traps)
10/29	(10) How to build an effective team
11/5	(11) Midterm <u>(Final Project Pitch)</u>
11/12	(12) Corporate Entrepreneurship
11/19	(13) Fundraising Strategy (Term Sheet Exercise)
11/26	(14) Startup Case Study (current trends)
12/3	(15) Final (Oral Presentations)
12/10	(16) Finals Week (written report due)